

Computer Science & IT

Discrete Mathematics

Mathematical Logic

Lecture No. 08

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Recap of Previous Lecture



Topic

Rules of inferences with respect to predicate logic



Topic

Important logical implications and equivalences



Topic

Practice questions



Topics to be Covered



Topic

Practice questions



Topic

Tautology w.r.t. predicate formula





Topic : Equivalences & Implications



⑥

$$\forall x [P(x) \leftrightarrow Q(x)] \not\Rightarrow (\forall x [P(x)] \leftrightarrow \forall x [Q(x)])$$

✓ 1. $\forall x [P(x) \wedge Q(x)] \equiv [\forall x P(x)] \wedge [\forall x Q(x)]$

✓ 2. $\exists x [P(x) \vee Q(x)] \equiv [\exists x P(x)] \vee [\exists x Q(x)]$

✓ 3. $[\forall x P(x)] \vee [\forall x Q(x)] \not\Rightarrow \forall x [P(x) \vee Q(x)]$

✓ 4. $\exists x [P(x) \wedge Q(x)] \not\Rightarrow [\exists x P(x)] \wedge [\exists x Q(x)]$

✓ 5. $\forall x [P(x) \rightarrow Q(x)] \not\Rightarrow [\forall x P(x)] \rightarrow [\forall x Q(x)]$



Topic : Some important equivalences

Note:- Here Q is a Predicate formula without any free occurrence of ' x '
 $\Downarrow$
ie, truth value of Q does not vary with ' x '

$\Downarrow$
{ if Q is true then it will be True w.r.t. all x in domain

{ if Q is false, then it will remain false w.r.t. all x in domain

1. $\forall x [P(x) \wedge Q] \equiv \forall x P(x) \wedge Q$

2. $\exists x [P(x) \vee Q] \equiv \exists x P(x) \vee Q$

3. $\forall x [P(x) \vee Q] \equiv \forall x P(x) \vee Q$

4. $\exists x [P(x) \wedge Q] \equiv \exists x P(x) \wedge Q$



Topic : Some important equivalences

1. $\forall x [P(x) \wedge Q] \equiv \forall x P(x) \wedge Q$
 - Case ① When $Q = \text{False}$
if $Q = \text{False}$, then both LHS & R.H.S.
will be false for all values of x
i.e. same truth value of LHS & R.H.S.
 - Case ② When $Q = \text{True}$
(i) For LHS to be true, $P(x)$ must be true
for all x . And in that case $\forall x \{P(x)\}$ will
also be true, hence RHS will also be true
(ii) For RHS to be true, $\forall x \{P(x)\}$ must be true
i.e. $P(x)$ must be true for all x , in
that case LHS will also be true
2. $\exists x [P(x) \vee Q] \equiv \exists x P(x) \vee Q$
3. $\forall x [P(x) \vee Q] \equiv \forall x P(x) \vee Q$
4. $\exists x [P(x) \wedge Q] \equiv \exists x P(x) \wedge Q$



Topic : Some important equivalences

1. $\forall x [P(x) \wedge Q] \equiv \forall x P(x) \wedge Q$

2. $\exists x [P(x) \vee Q] \equiv \exists x P(x) \vee Q$

3. $\forall x [P(x) \vee Q] \equiv \forall x P(x) \vee Q$

4. $\exists x [P(x) \wedge Q] \equiv \exists x P(x) \wedge Q$

Similar logic can be analyzed
w.r.t. other three equivalences
as well.



Topic : Some important equivalences

5. $\forall x [P \rightarrow Q(x)] \equiv P \rightarrow \forall x Q(x)$

6. $\exists x [P \rightarrow Q(x)] \equiv P \rightarrow \exists x Q(x)$

Here 'P' is the predicate formula without any free occurrence of 'x'

7. $\forall x [P(x) \rightarrow Q] \equiv \exists x P(x) \rightarrow Q$

8. $\exists x [P(x) \rightarrow Q] \equiv \forall x P(x) \rightarrow Q$

Here, Q is the predicate formula without any free occurrence of x



Topic : Some important equivalences

$$\begin{aligned} 5. \quad & \forall x [P \rightarrow Q(x)] \equiv P \rightarrow \forall x Q(x) \\ & \equiv \forall x \{ \sim P \vee Q(x) \} \\ & \equiv \sim P \vee \forall x \{ Q(x) \} \equiv P \rightarrow \forall x \{ Q(x) \} = \text{RHS} \end{aligned}$$

$$\begin{aligned} 6. \quad & \exists x [P \rightarrow Q(x)] \equiv P \rightarrow \exists x Q(x) \\ & \equiv \exists x \{ \sim P \vee Q(x) \} \\ & \equiv \sim P \vee \exists x \{ Q(x) \} \equiv P \rightarrow \exists x \{ Q(x) \} = \text{RHS} \end{aligned}$$

$$7. \quad \forall x [P(x) \rightarrow Q] \equiv \exists x P(x) \rightarrow Q$$

$$8. \quad \exists x [P(x) \rightarrow Q] \equiv \forall x P(x) \rightarrow Q$$



Topic : Some important equivalences

5. $\forall x [P \rightarrow Q(x)] \equiv P \rightarrow \forall x Q(x)$

6. $\exists x [P \rightarrow Q(x)] \equiv P \rightarrow \exists x Q(x)$

7. $\forall x [P(x) \rightarrow Q] \equiv \exists x P(x) \rightarrow Q$

$$\begin{aligned} \forall x \{(\sim P(x)) \vee Q\} &\equiv \forall x \{ \underbrace{\sim P(x)}_{\sim \exists x P(x)} \vee Q \} \equiv \sim(\exists x \{P(x)\}) \vee Q \\ &\equiv \exists x \{P(x)\} \rightarrow Q \equiv \text{R.H.S.} \end{aligned}$$

8. $\exists x [P(x) \rightarrow Q] \equiv \forall x P(x) \rightarrow Q$

$$\begin{aligned} \equiv \exists x \{ \underbrace{\sim P(x)}_{\sim \forall x P(x)} \vee Q \} &\equiv \exists x \{ \sim P(x) \vee Q \} \equiv \sim(\forall x \{P(x)\}) \vee Q \equiv \forall x \{P(x)\} \rightarrow Q \equiv \text{R.H.S.} \end{aligned}$$

#Q. Let $P(x)$ and $Q(x)$ be arbitrary predicates. Which of the following statements is always TRUE?

~~A~~

$$(\forall x(P(x) \vee Q(x))) \not\Rightarrow ((\forall x P(x)) \vee (\forall x Q(x)))$$

← ✓

~~B~~

$$(\forall x(P(x) \Rightarrow Q(x))) \not\Rightarrow ((\forall x P(x)) \Rightarrow (\forall x Q(x)))$$

← ✗

~~C~~

$$(\forall x(P(x) \Rightarrow \forall x Q(x))) \not\Rightarrow (\forall x (P(x) \Rightarrow Q(x)))$$

← ✓

~~D~~

$$((\forall x(P(x) \Leftrightarrow \forall x Q(x))) \not\Rightarrow (\forall x(P(x) \Leftrightarrow Q(x))))$$

← ✓

#Q. Which of the following predicate calculus statements is/are valid Let $U = \{1, 2\}$

☒ **A**

$$(\forall x)P(x) \vee (\forall x)Q(x) \rightarrow (\forall x)\{P(x) \vee Q(x)\}$$

☒ **B**

$$(\exists x)P(x) \wedge (\exists x)Q(x) \rightarrow (\exists x)\{P(x) \wedge Q(x)\}$$

☒ **C**

$$(\exists x)\{P(x) \vee Q(x)\} \rightarrow (\forall x)P(x) \vee (\forall x)Q(x)$$

$$\checkmark \equiv \exists x\{P(x)\} \vee \exists x\{Q(x)\}$$

☒ **D**

$$(\exists x)\{P(x) \vee Q(x)\} \rightarrow \{\exists x P(x) \vee \forall x Q(x)\}$$

$$\equiv$$

$$\exists x$$

let $P = \text{False for all } x$
 $\& Q(1) = T \& Q(2) = F$

$$\therefore \begin{matrix} P(1) & \vee & Q(1) & = & T \\ F & & T & & \end{matrix}$$

$\&$

$$\begin{matrix} P(2) & \vee & Q(2) & = & F \\ F & & F & & \end{matrix}$$

$\therefore \exists x\{P(x) \vee Q(x)\}$
 is true
 So LHS is true

but $P = \text{False for all } \therefore \exists x\{P(x)\} = F$
 $Q(2) = F \therefore \forall x\{Q(x)\} = F$
 Hence $\exists x\{P(x)\} \vee \forall x\{Q(x)\} = F \vee F = F$

Today's Topic

#Q. Which of the following predicate calculus statements is/are valid

☒ A

$$\forall(x)(P(x) \Rightarrow Q(x)) \Rightarrow ((\forall(x)P(x)) \Rightarrow (\forall(x)Q(x)))$$

☒ B

$U = \{1\} \quad P(1) = T \quad Q(1) = F$

$$\exists x(P(x) \vee Q(x)) \Rightarrow ((\exists xP(x)) \Rightarrow (\exists xQ(x)))$$

$P(1) \vee Q(1) = T \quad T \rightarrow F = F$

☒ C

$$\exists x(P(x) \wedge Q(x)) \Leftrightarrow ((\exists xP(x)) \wedge (\exists xQ(x)))$$

☒ D

$$\forall(x)\exists y P(x, y) \Leftrightarrow \exists y \forall(x)P(x, y)$$

H.W.

#Q. Which of the following is a valid first order formula? (Here α and β are first order formulae with x as their only free variable)

☒ A

$$((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]) \Rightarrow (\forall x)[\alpha \Rightarrow \beta]$$

$$= (\forall x \{P(x) \rightarrow Q(x)\}) \rightarrow \forall x \{P(x) \rightarrow Q(x)\}$$

☒ B

$$((\forall x)[\alpha] \Rightarrow (\exists x)[\alpha \wedge \beta])$$

→ let α = True for all but β = False for all, then LHS = True & R.H.S = False

☒ C

$$(\forall x)[\alpha \vee \beta] \Rightarrow ((\exists x)[\alpha] \Rightarrow (\forall x)[\alpha])$$

$$\forall x \{P(x) \vee Q(x)\} \Rightarrow (\exists x \{P(x)\} \rightarrow \forall x \{P(x)\})$$

$$\sim (\exists x(P(x)) \vee \forall x P(x))$$

$$\forall x \{ \sim P(x) \} \vee \forall x \{ P(x) \}$$

☒ D

$$((\forall x)[\alpha \Rightarrow \beta] \Rightarrow ((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]))$$

[MSQ]



H.W.

#Q. Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first-order formula with x as a free variable and β is a first-order formula with no free variable.



$$[\beta \rightarrow (\exists x, \alpha(x))] \rightarrow [\forall x, \beta \rightarrow \alpha(x)]$$

valid w.r.t. non-empty domain



$$[\exists x, \beta \rightarrow (\alpha(x))] \rightarrow [\beta \rightarrow (\forall x, \alpha(x))]$$

$$\exists x \{ \sim \beta \vee \alpha(x) \}$$

$$\sim \beta \vee \exists x \{ \alpha(x) \}$$

$$\sim \beta \vee \forall x \{ \alpha(x) \}$$



$$[(\exists x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$$



$$[(\forall x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$$

$$\sim (\forall x \{ \alpha(x) \}) \vee \beta$$

$$\exists x \{ \sim \alpha(x) \} \vee \beta$$

$$\forall x \{ \sim \alpha(x) \vee \beta \} \equiv \forall x \{ \sim \alpha(x) \} \vee \beta$$

at least one 'x'

$$\beta \rightarrow \exists x \{ \alpha(x) \} \rightarrow \exists x \{ \beta \rightarrow \alpha(x) \}$$

$$(\beta \rightarrow \exists x \{ \alpha(x) \}) \rightarrow \forall x \{ \beta \rightarrow \alpha(x) \}$$

$\alpha(x)$ must be true for at least one 'x'

for it to be true α must be true for all

H.W.

#Q. Which one of the following predicate formulae is NOT logically valid? Note that W is a predicate formula without any free occurrence of x .

- A** $\exists x(p(x) \wedge W) \equiv (\exists x p(x)) \wedge W$ { valid }
- B** $\forall x(p(x) \rightarrow W) \equiv (\forall x p(x)) \rightarrow W$ { invalid }
 $\equiv \exists x \{p(x)\} \rightarrow W$
- C** $\exists x(p(x) \rightarrow W) \equiv (\forall x p(x)) \rightarrow W$ { valid }
- D** $\forall x(p(x) \vee W) \equiv (\forall x p(x)) \vee W$ { valid }

#Q. What is the correct translation of the following statement into mathematical logic?

at least one number is real and rational
 "some real numbers are rational"

- A** $\exists x (real(x) \vee rational(x))$ { some number are real or rational }
- B** $\exists x (real(x) \wedge rational(x))$ { some numbers are real and rational }
- C** $\forall x (real(x) \rightarrow rational(x))$ { for all x , if ' x ' is real then x is rational }
 All real numbers are rational }
- D** $\forall x (rational(x) \rightarrow real(x))$ { All rational numbers are real }

In general,

w.r.t. "some" we use ' $\exists$ '

and w.r.t. "all" we use ' $\forall$ '

Note eg: All are human and intelligent
(Not always)

$$\equiv \forall x \{ H(x) \wedge I(x) \}$$

{ Let $H(x)$ = x is human
 $I(x)$ = x is intelligent

eg: All human are intelligent

$$\equiv \forall x \{ H(x) \rightarrow I(x) \}$$

[MSQ]

#Q. What is the logical translation of the following statement?

All my friends are imperfect

$$\forall x \{ F(x) \rightarrow \sim P(x) \}$$



let, $\sim F(x) \equiv x$ is enemy

"None of my friends are perfect"

$\left. \begin{array}{l} F(x): x \text{ is friend} \\ P(x): x \text{ is perfect} \end{array} \right\}$

Not at least one of my friend is perfect
 $\exists x (F(x) \wedge P(x))$

A

$$\exists x (F(x) \wedge \sim P(x))$$

$\equiv$ Some of my friends are not perfect

B

$$\exists x (\sim F(x) \wedge \sim P(x))$$

$\equiv$ Some of my enemy are not perfect

C

$$\exists x (\sim F(x) \wedge P(x))$$

$\equiv$ Some of my enemy are perfect

D

$$\sim (\exists x (F(x) \wedge P(x)))$$

$$\sim \exists x \{ F(x) \wedge P(x) \}$$

$$\equiv \forall x \{ \sim (F(x) \wedge P(x)) \}$$

$$\equiv \forall x \{ \sim F(x) \vee \sim P(x) \}$$

$$\equiv \forall x \{ F(x) \rightarrow \sim P(x) \}$$

H.W.
[MSQ]

#Q. The CORRECT formula of the sentence "not all rainy days are cold"

- A** $\forall d (\text{rainy}(d) \wedge \sim \text{cold}(d))$
- B** $\exists d (\sim \text{rainy}(d) \rightarrow \text{cold}(d))$
- C** $\forall d (\sim \text{rainy}(d) \rightarrow \text{cold}(d))$
- ☒ **D** $\exists d (\text{rainy}(d) \wedge \sim \text{cold}(d))$

Some rainy days are not cold
 $\exists x \{ \text{rainy}(x) \wedge \sim \text{cold}(x) \}$

$\sim (\forall x (\text{rainy}(x) \rightarrow \text{cold}(x)))$

$\sim (\forall x (\sim \text{rainy}(x) \vee \text{cold}(x)))$

$\exists x \{ \text{rainy}(x) \wedge \sim \text{cold}(x) \}$

PW

[MSQ]

H.W.

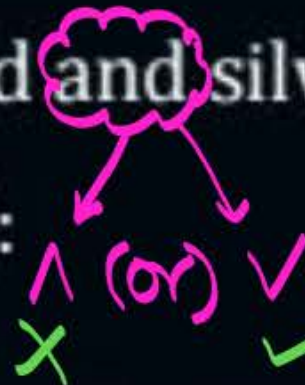
#Q.

Which one the following is the most appropriate logical formula to represent the statement "gold and silver ornaments are precious". The following notations are used:

$S(x)$: x is a silver ornament

$G(x)$: x is a gold ornament

$P(x)$: x is precious



A

$$\forall x (P(x) \rightarrow G(x) \wedge S(x))$$

X

B

$$\exists x ((G(x) \wedge S(x)) \rightarrow P(x))$$

X

C

$$\forall x ((G(x) \wedge S(x)) \rightarrow P(x))$$

X

D

$$\forall x ((G(x) \vee S(x)) \rightarrow P(x))$$

$\forall x$ { if ornament x is of gold or silver then that x is precious }

H.W. [MSQ]



#Q. Which one of the first order predicate calculus statements given below correctly expresses the following English statements? "Tigers and lions attack if they are hungry or threatened".

general statement

(if they are "hungry or threatened" then (attack))

A

$\forall x[(tiger(x) \wedge lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \rightarrow attacks(x))]$

~~**B**~~

$\forall x[(tiger(x) \vee lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \wedge attacks(x))]$

C

$\forall x[(tiger(x) \wedge lion(x)) \rightarrow (attacks(x) \rightarrow (hungry(x) \vee threatened(x)))]$

D

$\forall x[(tiger(x) \vee lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \rightarrow attacks(x))]$

H.W.
#Q.

Which of the following argument is ^{are} valid?

- A** $\forall x (P(x))$ follows from the premises $\exists x (P(x) \wedge Q(x))$
- B** $\exists x (P(x) \wedge Q(x))$ follows from the premises $\exists x (P(x)) \wedge \exists x (Q(x))$
- C** $\exists x (Q(x))$ follows from the premises $\{\forall x (P(x) \Rightarrow Q(x)), \exists y (Q(y))\}$
- D** $\sim P(a, b)$ follows from the premises $\{\forall x \forall y \{ P(x, y) \rightarrow W(x, y) \}, \sim W(a, b)\}$

H.W.
#Q.

Check whether the following argument is true or false?

$\forall x \{F(x) \rightarrow \sim S(x)\}$ follows from the premises

$\{ \exists x(F(x) \wedge S(x)) \rightarrow \forall y(M(y) \rightarrow W(y)) , \exists y(M(y) \wedge \sim W(y)) \}$



Topic : Tautology in predicate logic

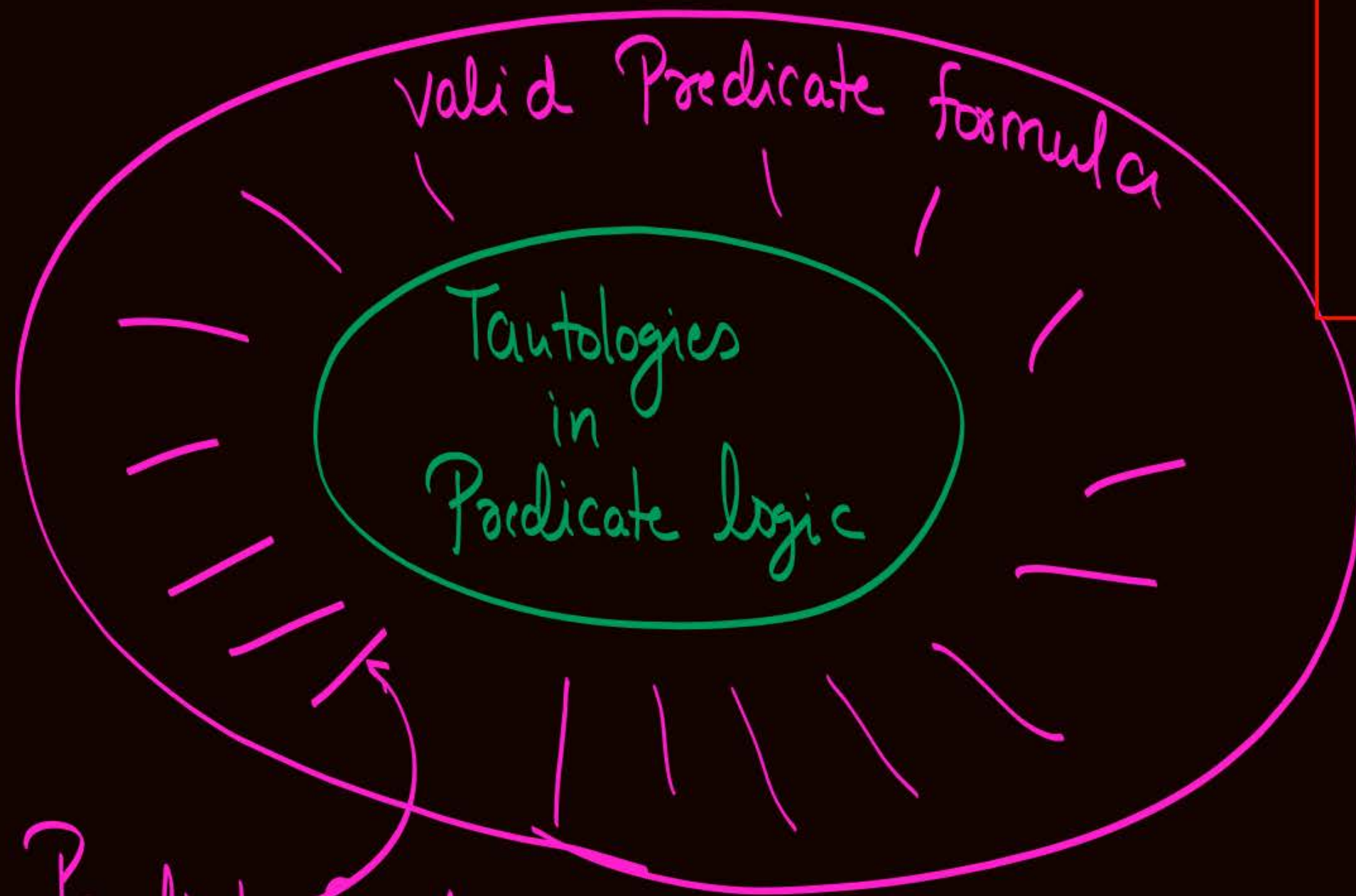
- ✦ In propositional logic, tautologies and validities are same.
- ✦ A propositional function which is always true is called valid propositional function or tautology.

But in predicate logic, distinction is maintained between logical validities and tautologies.

A predicate formula which is always true is called a valid predicate formula but it may or may not be a tautology. In predicate logic tautologies are proper subset of logical validities.

Tautologies are defined w.r.t. Propositional formula but if tautology then valid

W.r.t. Predicate,
if valid then need not be tautology
but if tautology then valid



Predicate formula
that are valid, but not tautology

Note: If a predicate formula is invalid then it can never be a tautology

Every tautology in predicate logic is a valid predicate formula, but every valid predicate formula need not be a tautology.



Topic : Tautology in predicate logic

$P \rightarrow (Q \rightarrow P)$ is tautology
let $P = \forall x \{P(x)\}$ & $Q = \exists x \{Q(x)\}$
then $\Rightarrow \forall x \{P(x)\} \rightarrow \{ \exists x \{Q(x)\} \rightarrow \forall x \{P(x)\} \}$ is a tautology

A tautology in predicate logic is a sentence that can be obtained by taking a tautology of propositional logic and uniformly replacing each propositional variable by a predicate formula (one formula per propositional variable).

NOTE: To check whether a given predicate formula is a tautology or not we can only use the concepts that we have learned in propositional logic,

{ i.e. we can not use any concept from Predicate logic }
Because tautologies are defined w.r.t. Propositional logic

Q: Which of the following predicate formula is/are tautology

(a) $\exists x \{P(x) \wedge Q(x)\} \implies (\exists x \{P(x)\} \wedge \exists x \{Q(x)\})$

(b) $\exists x \{P(x) \rightarrow Q(x)\} \implies \exists x \{\sim P(x) \vee Q(x)\}$

(c) $\sim \forall x \{P(x) \wedge Q(x)\} \iff \exists x \{P(x) \rightarrow \sim Q(x)\}$

(d) $\exists x \forall y (P(x, y)) \implies \forall y \exists x (P(x, y))$

Q: Which of the following predicate formula is/are tautology

$$\textcircled{a} \exists x \{P(x) \wedge Q(x)\} \implies (\exists x \{P(x)\} \wedge \exists x \{Q(x)\})$$

We know it is a valid Predicate formula

let $\exists x \{P(x) \wedge Q(x)\} \equiv A$ } then formula takes the form

$$A \longrightarrow (b \wedge c)$$

$$\exists x \{P(x)\} \equiv b$$

$$\exists x \{Q(x)\} \equiv c$$

And, it is not a tautology
{ i. given formula is valid }
but not a tautology

Q: Which of the following Predicate formula is/are tautology

(b) $\exists x \{P(x) \rightarrow Q(x)\} \Rightarrow \exists x \{\sim P(x) \vee Q(x)\}$ } it is valid as well as a tautology

it is a valid Predicate formula

from propositional logic we know $P \rightarrow Q \equiv \sim P \vee Q$

$\therefore P(x) \rightarrow Q(x) \equiv \sim P(x) \vee Q(x)$

$\therefore \exists x \{P(x) \rightarrow Q(x)\} \equiv \exists x \{\sim P(x) \vee Q(x)\}$

Wrt. Propositional logic as well

let $A = \exists x \{P(x) \rightarrow Q(x)\}$

$\therefore \exists x \{\sim P(x) \vee Q(x)\}$
is also $\equiv A$

$\therefore$ formula takes the form

$A \rightarrow A$ and it is a tautology
Hence given formula is a tautology

Q: Which of the following predicate formula is/are tautology

$$\textcircled{c} \sim \forall x \{ P(x) \wedge Q(x) \} \iff \exists x \{ P(x) \longrightarrow \sim Q(x) \}$$

it is a valid predicate formula

$$\sim \forall x \equiv \exists x$$

we know it from Predicate logic, not from propositional logic

∴ W.r.t. Propositional logic $\sim \forall x \{ P(x) \wedge Q(x) \} \neq \exists x \{ P(x) \longrightarrow \sim Q(x) \}$

∴ we must assign different variables

let $a \equiv \sim \forall x \{ P(x) \wedge Q(x) \}$ } formula becomes

& $b \equiv \exists x \{ P(x) \longrightarrow \sim Q(x) \}$ } $a \iff b$

it is not a tautology

∴ given formula is Valid, but not a tautology

Q: Which of the following predicate formula is/are tautology

(d) $\exists x \forall y (P(x,y)) \implies \forall y \exists x (P(x,y))$

it is valid predicate formula

Wrt. Propositional logic

$$\exists x \forall y \neq \forall y \exists x$$

$\therefore$ let $\left. \begin{array}{l} \exists x \forall y (P(x,y)) = a \\ \forall y \exists x (P(x,y)) = b \end{array} \right\} \implies \text{formula becomes } a \longrightarrow b$
it is not a tautology

$\therefore$ given formula is Valid but not a tautology.

Q: Which of the following predicate formula is/are tautology

~~(a)~~ $\exists x \{P(x) \wedge Q(x)\} \implies \exists x \{P(x)\} \wedge \exists x \{Q(x)\}$ { valid only }

☒ (b) $\exists x \{P(x) \rightarrow Q(x)\} \implies \exists x \{\sim P(x) \vee Q(x)\}$ { valid + Tautology }

~~(c)~~ $\sim \forall x \{P(x) \wedge Q(x)\} \iff \exists x \{P(x) \rightarrow \sim Q(x)\}$ { valid only }

~~(d)~~ $\exists x \forall y (P(x,y)) \implies \forall y \exists x (P(x,y))$ { valid only }



2 mins Summary



✓ **Topic**

Practice questions

✓ **Topic**

Tautology w.r.t. predicate formula

THANK - YOU